

READING PASSAGE 2

Nanotechnology Applications in Medicine

A The study of nanotechnology applications in medicine has undergone a remarkable transformation over the past several decades. What was once a niche area of academic inquiry has developed into a rich and diverse field of research, drawing on insights from a wide range of disciplines and theoretical traditions.

B The origins of contemporary research in this area can be traced to pioneering studies conducted during the mid-twentieth century. These early investigations, though constrained by the technological and methodological limitations of their era, established foundational principles that continue to inform current scholarship. Successive generations of researchers have built upon these foundations, progressively refining theoretical models and expanding the empirical evidence base through increasingly sophisticated research designs.

C The mechanisms underlying these phenomena operate at multiple levels of analysis, from the micro-level of individual decision-making to the macro-level of institutional and structural change. At the individual level, cognitive processes, emotional responses, and social influences shape the choices that people make. At the organisational level, formal rules, informal norms, and power dynamics govern collective action. At the societal level, historical legacies, cultural values, and political structures create the broad context within which lower-level processes unfold.

D The empirical record is further strengthened by a series of meta-analyses that have synthesised findings from hundreds of individual studies. These comprehensive reviews have consistently identified moderate to large effect sizes, with the most robust findings demonstrating remarkable consistency across different research designs, measurement approaches, and analytical strategies. This convergence of evidence from multiple independent lines of inquiry considerably strengthens confidence in the validity of the core theoretical propositions.

E The practical application of research findings has proven more challenging than many initially anticipated. The translation of academic knowledge into actionable policy recommendations requires navigating a complex landscape of political constraints, economic interests, and social dynamics that academic models often simplify or overlook. Implementation failures in several high-profile initiatives have underscored the gap between theoretical understanding and practical capacity, prompting calls for more engaged and context-sensitive approaches to research.

F Looking ahead, the field stands at a critical juncture. Emerging technologies offer unprecedented opportunities for data collection and analysis, potentially enabling researchers to address questions that have long remained beyond the reach of empirical investigation. At the same time, the accelerating pace of social and technological change means that the phenomena under study are themselves evolving, requiring theoretical frameworks that can accommodate dynamism and discontinuity rather than assuming stability and equilibrium.

G The evidence reviewed above demonstrates both the substantial achievements of research in this field and the significant work that remains to be done. As new tools and theoretical perspectives become available, and as the phenomena themselves continue to evolve, the research community will need to maintain both rigour in its methods and openness to new ideas. The stakes are high, for the questions addressed by this research bear directly on some of the most important challenges facing humanity in the twenty-first century.

Question 1

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 1 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.

Question 2

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 2 The author believes that all research questions in this field have been answered.

Question 3

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 3 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Question 4

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 4 The landmark study discussed in the passage involved participants from multiple countries.

Questions 5–7

Choose the correct letter, A, B, C or D.

- 5 The meta-analyses cited in the passage reveal that:
- A. pessimism about the future
 - B. cautious optimism with acknowledgment of challenges
 - C. complete certainty about outcomes
 - D. indifference to practical applications
- 6 According to the passage, what is one of the key factors identified by researchers?
- A. excessive research funding
 - B. translating academic findings into practical policy
 - C. lack of public interest
 - D. too many international collaborations
- 7 What does the passage identify as a significant challenge to current understanding?
- A. excessive funding
 - B. theoretical and methodological limitations

- C. public interest
- D. international collaboration

Question 8

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 8 According to the text, cognitive processes shape the _____ that people make.

Question 9

Choose the correct letter, A, B, C or D.

- 9 The author suggests that early studies in this field were limited by:
- A. excessive research funding
 - B. translating academic findings into practical policy
 - C. lack of public interest
 - D. too many international collaborations

Questions 10–11

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 10 The author believes that all research questions in this field have been answered.
- 11 Early researchers had access to the same technologies as modern scholars.

Question 12

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 12 The landmark study analysed data spanning a period of _____ years.

Question 13

Choose the correct letter, A, B, C or D.

- 13 What finding emerged from the landmark study discussed in the passage?
- A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations